

AI FITNESS & WELLNESS PLATFORM

Master Product Scope & Development Blueprint

1. Product Vision

Build a comprehensive **AI-powered Fitness & Wellness Platform** for:

- Fitness centers
- Gyms
- Fitness studios
- Personal trainers
- Online coaches
- Nutrition coaches
- Multi-location fitness businesses
- Individual fitness consumers

The platform should support both:

1. **Offline fitness-center operations**
2. **Online fitness and coaching**

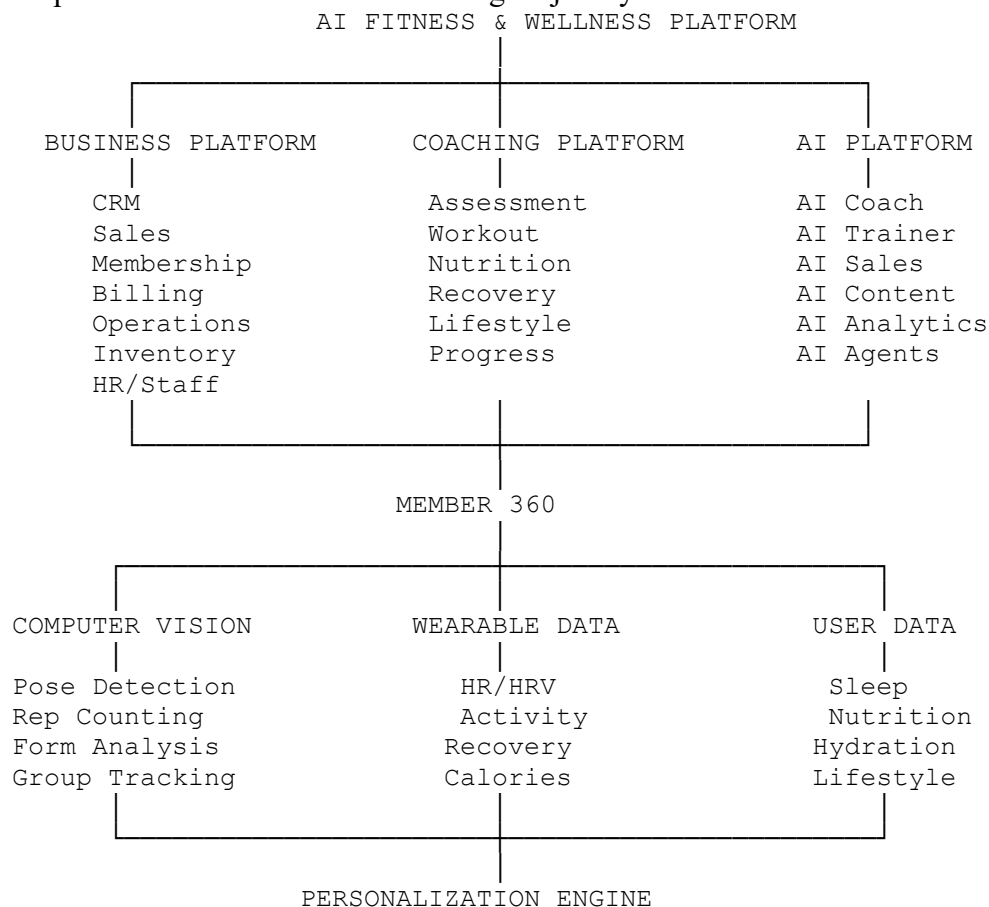
The system should combine:

CRM + Gym Management + Personal Training + Online Coaching + AI Coach + Nutrition + Body Assessment + Computer Vision + Customer Engagement + Commerce + Analytics

The platform should be designed as a **multi-tenant, API-first, modular architecture** so that additional fitness centers, trainers, brands and white-label deployments can be added later.

2. Core Product Architecture

The platform should have the following major layers:



3. User Roles

The system should support role-based access control.

Platform Level

- Super Admin
- Platform Administrator
- Platform Support
- Platform Finance
- Platform AI/ML Administrator

Business Level

- Business Owner
- Business Admin
- Center Manager
- Finance Manager
- HR/Admin
- Sales Manager
- Marketing Manager
- Operations Manager

Coaching Level

- Head Trainer
- Personal Trainer
- Group Trainer
- Online Coach
- Nutritionist
- Wellness Coach

Customer Level

- Member
- Online Client
- Guest
- Trial User

Support Level

- Customer Success
- Grievance Officer
- Support Executive

4. Multi-Tenant & Organization Management

The platform must support multiple organizations.

Each organization can have:

- Multiple branches
- Multiple gyms/studios
- Multiple trainers
- Multiple memberships
- Multiple programs
- Multiple pricing plans
- Multiple staff
- Multiple inventory locations

Tenant isolation must be implemented at application and database level.

Configuration should include:

- Branding
 - Logo
 - Theme
 - Domain/subdomain
 - Currency
 - Tax configuration
 - Membership rules
 - Attendance rules
 - Cancellation rules
 - Notification rules
 - AI policies
 - Consent policies
 - Data retention rules
-

5. Member 360

This should be the central entity of the entire platform.
Every member should have one unified profile.

Member Profile

- Personal information
- Contact details
- Profile photograph
- Emergency contact
- Fitness goals
- Experience level
- Preferences
- Membership
- Trainer
- Program
- Assessment
- Body measurements
- Workout history
- Nutrition
- Sleep
- Hydration
- Activity
- Wearables
- Attendance
- Payments
- Purchases
- Complaints
- Feedback
- Communication
- AI recommendations
- Progress
- Consent
- Data access history

Member Timeline

Everything should appear in a chronological timeline:

Registration

↓

Assessment

↓

Membership

↓

Workout

↓

Trainer Interaction

↓

Nutrition

↓

Purchase

↓

Complaint

↓

AI Recommendation

↓

Progress

This becomes the foundation for AI personalization.

6. CRM

A complete fitness CRM should be included.

Lead Management

- Lead capture
- Website leads
- App leads
- Walk-ins
- Referral leads
- Campaign leads
- Social media leads
- WhatsApp leads
- Imported leads

Lead Pipeline

Example:

New

↓

Contacted

↓

Qualified

↓

Assessment Booked

↓

Assessment Completed

↓

Trial

↓

Proposal

↓

Converted

↓

Membership Active

CRM Features

- Lead scoring
 - Lead assignment
 - Follow-up
 - Tasks
 - Reminders
 - Calls
 - WhatsApp
 - Email
 - Campaigns
 - Source tracking
 - Conversion tracking
 - Lost-lead reasons
 - Referral management
-

7. AI Sales Agent

AI should assist the sales team.

AI capabilities

- Lead qualification
- Goal discovery
- Fitness requirement discovery
- Package recommendation
- Objection handling
- Follow-up
- Appointment booking
- Assessment booking
- Renewal reminders
- Upsell recommendations

Channels

- Website chat
- WhatsApp
- Mobile app
- Voice
- Email

AI Calling

Integrate a voice AI layer such as Sarvam or another approved provider.

AI calling should:

- Identify lead
- Understand requirement
- Ask qualification questions
- Schedule appointment
- Follow up
- Update CRM
- Generate call summary
- Identify buying intent
- Escalate to human sales staff

AI should **never make unauthorized commitments regarding medical or health outcomes.**

8. Membership Management

Support:

- Membership plans
- Monthly plans
- Quarterly plans
- Annual plans
- Personal training packages
- Class packages
- Online coaching
- Hybrid plans
- Trial plans
- Family plans
- Corporate plans
- Promotional plans

Membership Features

- Start/end date
 - Freeze
 - Extension
 - Upgrade
 - Downgrade
 - Transfer
 - Cancellation
 - Renewal
 - Auto-renewal
 - Discount
 - Coupon
 - Referral benefit
-

9. Class & Schedule Management

Support:

- Group classes
- Personal training
- Online classes
- Hybrid classes
- Workshops
- Events
- Bootcamps
- Challenges

Features

- Class creation
- Trainer allocation
- Room allocation
- Capacity
- Waitlist
- Booking
- Cancellation
- Rescheduling
- Attendance
- Recurring classes

10. Trainer Management

Trainer profile should include:

- Qualifications
- Certifications
- Specializations
- Experience
- Availability
- Assigned members
- Classes
- Performance
- Revenue generated
- Member ratings

Trainer Dashboard

Trainer should see:

- Today's sessions
- Assigned members
- Member progress
- Missed sessions
- Members requiring attention
- AI recommendations
- Program reviews
- Upcoming renewals
- Complaints
- Performance

11. Fitness Assessment

This is a major module.

Assessment Types

- Initial assessment
- Periodic assessment
- Fitness assessment
- Mobility assessment
- Strength assessment
- Cardio assessment
- Flexibility
- Balance
- Functional movement
- Body measurement
- Posture assessment

Body Data

- Height
- Weight
- Waist
- Hip
- Chest
- Arm
- Thigh
- BMI

- Body-fat estimate where supported
- Muscle-mass estimate where supported

Avoid presenting estimates as clinical measurements unless appropriately validated.

12. AI Body Assessment

Allow:

- Camera-based assessment
- Image-based analysis
- Posture analysis
- Body measurement estimation
- Movement assessment
- Progress comparison

Before/After

Member can compare:

- Photos
- Measurements
- Fitness scores
- Workout performance
- Weight trends
- Body composition estimates

AI Progress Analysis

AI should explain:

- What improved
 - What declined
 - What remained unchanged
 - Where to focus next
-

13. Fitness Score

Create a configurable fitness score.

Example:

Overall Fitness Score	78
Strength	74
Cardio	82
Mobility	68
Flexibility	71
Consistency	89
Recovery	73
Nutrition	76
Movement Quality	81

The scoring formula should be configurable rather than hard-coded.

14. Exercise Library

Create a central exercise database.

Each exercise should contain:

- Exercise name
- Category
- Muscle groups
- Equipment
- Difficulty

- Instructions
- Video
- Images
- Contraindications/warnings
- Required movement pattern
- Sets
- Reps
- Duration
- Tempo
- Rest
- Variations
- Progressions
- Regressions

15. Exercise Content Generator

AI should generate exercise content.

Input:

Goal

Fitness level

Equipment

Duration

Age band

Training frequency

Preferences

Assessment

Previous performance

Output:

- Exercise selection
- Instructions
- Sets
- Reps
- Tempo
- Rest
- Progression
- Alternatives

Trainer must be able to review and modify AI-generated content.

16. Workout Program Builder

Support:

- Single workouts
- Weekly programs
- Monthly programs
- Multi-week programs
- Challenges
- Group programs
- Personal training programs
- Online coaching programs

Program Logic

Goal

↓

Assessment

↓
Fitness Level
↓
Program
↓
Workout
↓
Execution
↓
Performance
↓
Adjustment

17. AI Workout Generator

AI should generate personalized programs based on:

- Goal
- Assessment
- Available equipment
- Schedule
- Training history
- Performance
- Recovery
- Preferences

The AI must explain **why** the workout was selected.

18. Nutrition & Diet Management

Nutrition Profile

- Goal
- Food preferences
- Dietary preferences
- Meal frequency
- Allergies
- Restrictions
- Calorie target
- Macro targets

Diet Generator

Generate:

- Daily meal plan
- Weekly meal plan
- Recipes
- Food alternatives
- Shopping list
- Calories
- Protein
- Carbohydrates
- Fat
- Micronutrient information where reliable

Indian Food Database

Include Indian food items and regional variations.

Support:

- Vegetarian

- Non-vegetarian
- Vegan
- Jain
- Regional diets
- Indian meal patterns

Nutrition recommendations should include appropriate disclaimers and escalation to qualified professionals for medical/disease-specific diets.

19. Supplement Management

Support:

- Supplement catalogue
- Product information
- Inventory
- Recommendations
- Purchase
- Subscription
- Consumption tracking

AI should distinguish between:

general fitness guidance

and

medical/supplement advice requiring professional review.

20. Lifestyle & Wellness

Sleep

- Manual logging
- Wearable integration
- Sleep duration
- Sleep consistency
- Sleep quality
- Recovery indicators

Hydration

- Water logging
- Daily targets
- Reminder
- Activity-based recommendation

Activity

- Steps
- Active minutes
- Calories
- Sedentary periods

Lifestyle

- Stress self-report
 - Energy
 - Mood
 - Daily habits
 - Recovery
-

21. AI Personal Coach

This is one of the flagship modules.

The AI Coach should understand the member's entire context.

Member can ask:

What should I train today?

I missed yesterday's workout.

What should I eat after training?

I slept badly last night.

Can I replace this exercise?

Why am I not progressing?

The AI should retrieve relevant Member 360 information before responding.

22. Trainer Copilot

Trainer receives AI assistance.

Example:

Today's Priority Members

HIGH ATTENTION

Member A

3 missed sessions

Progress plateau

Renewal in 14 days

Member B

Form deterioration

3 consecutive sessions

Member C

Low engagement

No app activity for 8 days

Trainer can ask:

"Show members whose progress has stalled."

"Create next week's plan for Member A."

"Which members need assessment?"

"Summarize my client portfolio."

23. Computer Vision Platform

This should be a dedicated AI/CV subsystem.

Core capabilities

- Person detection
 - Person tracking
 - Pose estimation
 - Exercise recognition
 - Rep counting
 - Range-of-motion estimation
 - Tempo detection
 - Movement quality
 - Form deviation
 - Exercise classification
-

24. Exercise Form Analysis

Start with predefined exercises.

Example:

Squat

Detect:

- Depth
- Knee alignment
- Hip movement
- Trunk angle
- Tempo
- Stability

Output:

Rep 08

Depth	Good
Knee Alignment	Attention
Tempo	Good
Stability	Good

Movement Score: 84%

25. Multi-Person Group Class Tracking

The platform should support multiple people.

Example:

Class: Functional Training

Participants: 24

Normal:	18
Attention:	4
High Attention:	2
Unidentified:	0

Trainer dashboard should show members requiring attention.

Do not attempt to infer sensitive medical conditions from camera footage.

26. Real-Time Feedback

Feedback can be:

Member

- Screen
- Audio
- Mobile notification
- Visual overlay

Trainer

- Dashboard
- Alerts
- Participant ranking
- Attention queue

Feedback must be configurable because excessive alerts will distract both trainer and member.

27. Online Coaching

Support:

- 1:1 video coaching
- Group video classes
- Recorded sessions
- Live classes
- Program-based coaching

Online AI

- Camera access
- Pose estimation
- Rep counting
- Form analysis
- Exercise identification
- Workout tracking

Where possible, perform CV processing on the user's device or nearby edge infrastructure to reduce privacy exposure and latency.

28. Wearable Integration

Design an abstraction layer so multiple wearable providers can be connected.

Potential categories:

- Smart watches
- Smart rings
- Fitness bands
- Heart-rate devices

Data may include:

- Heart rate
- Activity
- Steps
- Sleep
- Workout
- Calories
- HRV where available
- Recovery metrics where available

Do not make the platform dependent on one wearable provider.

29. Adaptive Coaching Engine

This is the intelligence layer connecting everything.

Input:

Assessment
+
Workout
+
Performance
+
CV
+
Sleep
+
Recovery
+
Nutrition

+
Attendance
+
Goals
Output:
Next Workout
Nutrition Guidance
Recovery Guidance
Difficulty Adjustment
Trainer Alert
Member Notification

Every AI recommendation should have:

- Reason
 - Source data
 - Confidence where appropriate
 - Timestamp
 - Model/version
 - Human approval status where required
-

30. AI Recommendation Engine

Create a centralized recommendation engine rather than implementing AI logic separately in every module.

Examples:

- Workout recommendation
 - Exercise replacement
 - Nutrition recommendation
 - Recovery recommendation
 - Membership recommendation
 - Upsell recommendation
 - Trainer action
 - Retention action
 - Engagement notification
-

31. AI Agent Platform

Eventually create multiple specialized agents.

Member Coach Agent

Fitness + lifestyle support.

Trainer Agent

Client management + program support.

Sales Agent

Lead qualification + follow-up.

Retention Agent

Churn detection + engagement.

Nutrition Agent

Food and nutrition support.

Operations Agent

Center operational queries.

Business Intelligence Agent

Owner/management analytics.

All agents should use a shared permission and data-access layer.

32. Grievance & Customer Success

Complete ticketing system:

- Complaint
- Feedback
- Suggestion
- Escalation
- Refund request
- Trainer complaint
- Facility complaint
- Service issue

Workflow

Complaint

↓

Categorization

↓

Assignment

↓

SLA

↓

Investigation

↓

Resolution

↓

Customer Confirmation

↓

Closure

↓

CSAT

AI can classify complaints and suggest routing, but final resolution authority should remain with humans for sensitive cases.

33. Marketing & Engagement

Support:

- Campaigns
- Email
- SMS
- WhatsApp
- Push notifications
- Offers
- Referral programs
- Loyalty
- Challenges
- Community

AI Personalization

Different members should receive different engagement based on behavior.

34. Community

Build community functionality:

- Challenges
- Leaderboards

- Achievements
- Badges
- Streaks
- Groups
- Events
- Trainer posts
- Member posts
- Announcements

Include moderation and reporting.

35. Gamification

Support:

- Points
- Levels
- Streaks
- Badges
- Challenges
- Leaderboards
- Rewards

Examples:

7-Day Workout Streak

10 Classes Completed

100K Steps

30-Day Consistency

36. E-Commerce

Support fitness-related commerce.

Products

- Supplements
- Food
- Apparel
- Equipment
- Accessories

Commerce

- Product catalogue
 - Cart
 - Orders
 - Payments
 - Coupons
 - Inventory
 - Returns
 - Delivery
 - Subscription products
-

37. Inventory & Supply Chain

Support:

- Purchase
- Vendor

- GRN
- Stock
- Stock transfer
- Consumption
- Batch
- Expiry
- Reorder level
- Damaged stock
- Returns
- Multi-location inventory

Particularly important for supplements and consumables.

38. Finance

Support:

- Membership billing
- Invoices
- Payments
- Refunds
- Discounts
- Taxes
- Revenue
- Expenses
- Trainer payouts
- Commission
- Package accounting
- Outstanding payments

Integrate external accounting software through APIs rather than trying to recreate a full accounting ERP initially.

39. HR & Staff

Support:

- Employee profile
 - Trainer profile
 - Attendance
 - Leave
 - Shift
 - Payroll integration
 - Certifications
 - Training
 - Performance
 - Targets
 - Commission
-

40. LMS / Knowledge Management

Create an internal learning system.

Content

- Trainer training
- SOPs

- Exercise methodology
- Nutrition guidelines
- Safety
- Customer handling
- Sales
- Compliance

Support:

- Video
- PDF
- Documents
- Quizzes
- Assessments
- Certification
- Expiry
- Renewal

41. Business Intelligence

Dashboard for owner:

Revenue

- Revenue
- MRR
- ARPU
- Membership revenue
- PT revenue
- Online revenue
- Product revenue

Sales

- Leads
- Conversion
- Cost per lead
- Assessment conversion
- Sales funnel

Customer

- Active members
- New members
- Churn
- Renewal
- Engagement

Trainers

- Sessions
- Clients
- Utilization
- Revenue
- Ratings

Operations

- Class occupancy
- Equipment utilization
- Attendance
- Capacity

42. AI Business Intelligence

Owner should be able to ask:

"Why did revenue decline?"

"Which trainer has the highest retention?"

"Which membership should we promote?"

"Which members are likely to churn?"

"Which classes are underutilized?"

"What are our best-performing lead sources?"

The AI should generate the answer from actual platform data and provide supporting metrics.

43. Digital Twin / Personal Fitness Model

Create a long-term data model representing each member.

The model should contain:

Identity

Goals

Fitness Level

Body Metrics

Movement

Workout

Nutrition

Sleep

Recovery

Behaviour

Attendance

Progress

Preferences

Trainer Feedback

AI Recommendations

Outcomes

The goal is not merely to store data.

The goal is to understand:

What interventions work best for this particular person?

44. Predictive Intelligence

Future models can include:

Churn prediction

Probability of membership churn.

Engagement prediction

Probability of becoming inactive.

Program adherence

Likelihood of completing program.

Plateau detection

Identify stalled progress.

Recovery recommendation

Estimate appropriate training intensity.

Training load intelligence

Identify sudden changes in training volume/intensity.

Important:

These should initially be positioned as **decision-support and wellness insights**, not medical diagnosis or guaranteed injury prediction.

45. Security & Privacy

Because the system can contain sensitive personal, health-related and biometric information, security must be designed from day one.

Security

- TLS
- Encryption at rest
- Strong authentication
- MFA
- RBAC
- Tenant isolation
- API authorization
- Audit logging
- Secure secrets management
- Backup
- Disaster recovery
- Vulnerability management
- Secure development lifecycle

Privacy

Support:

- Consent management
- Purpose-based processing
- Data minimization
- Data retention
- Data deletion
- Data export
- Access history
- Revocation of consent

For India, design with the **Digital Personal Data Protection Act, 2023 and applicable rules/regulations** in mind, and implement privacy controls that can also support international deployments.

Biometric/face data and camera-based processing should receive **explicit, purpose-specific consent** where required.

46. AI Governance

Every AI recommendation should be traceable.

Store:

User

Input Data

Recommendation

Model

Model Version

Prompt/Policy Version

Timestamp

Confidence

Trainer Approval

Final Action

Outcome

This becomes the **AI audit trail**.

47. Human-in-the-Loop

The system should distinguish between:

AI can automatically execute

- Reminders
- Notifications
- Low-risk content suggestions
- CRM updates
- Summaries

AI recommends; human approves

- Workout program changes
- Significant nutrition changes
- Sensitive member cases
- Trainer escalation
- High-impact customer decisions

Human only

- Medical diagnosis
- Medical treatment
- Injury diagnosis
- Emergency decisions
- Clinical interpretation

This distinction should be built into the permission architecture.

48. API & Integration Platform

Build API-first.

Potential integrations:

Communication

- WhatsApp
- SMS
- Email
- Push

Payments

- Indian payment gateways

AI

- LLM provider abstraction
- Voice AI
- Speech-to-text
- Text-to-speech

Computer Vision

- Pose models
- CV inference service

Wearables

- Apple Health
- Google Health Connect
- Wearable APIs where permitted

Business

- Accounting
- ERP

- Marketing platforms

Video

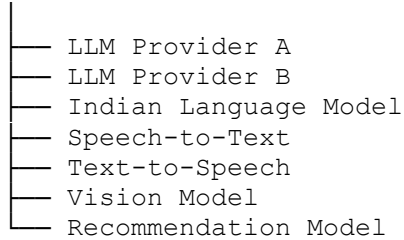
- WebRTC
- Video conferencing
- Streaming

49. AI Model Abstraction

Do not hard-code the platform to one AI provider.

Create:

AI Gateway



This allows the business to change models without rebuilding the application.

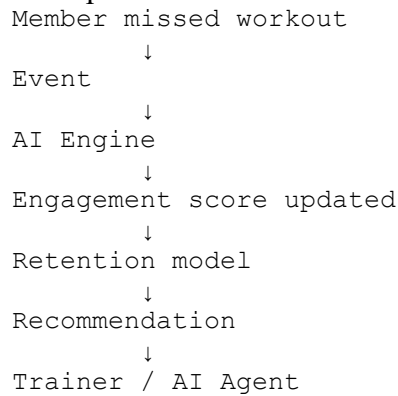
50. Data Architecture

The core data domains should include:

Tenant
 Organization
 Branch
 User
 Member
 Trainer
 Lead
 Membership
 Subscription
 Assessment
 BodyMetric
 Exercise
 Workout
 WorkoutSession
 NutritionPlan
 Meal
 Sleep
 Activity
 WearableData
 CVSession
 PoseMetric
 AIRecommendation
 AIConversation
 AIInteraction
 Complaint
 Ticket
 Payment
 Invoice
 Product
 Inventory
 Order
 Campaign
 Notification
 AuditLog
 Consent

Use a centralized event model so that important actions can become AI signals.

Example:



51. Event & Intelligence Layer

Create a common event architecture.

Events may include:

- Member registered
- Assessment completed
- Workout completed
- Workout missed
- Exercise completed
- Form issue detected
- Nutrition logged
- Sleep logged
- Member inactive
- Membership nearing expiry
- Complaint raised
- Payment failed
- Purchase completed

This event layer will be extremely important for future AI.

52. Mobile Applications

Member App

- Dashboard
- AI Coach
- Workout
- Nutrition
- Sleep
- Water
- Progress
- Booking
- Classes
- Trainer
- Chat
- Payments
- Store
- Challenges
- Notifications

Trainer App

- Today's schedule
 - Members
 - Workout
 - Assessment
 - Progress
 - AI recommendations
 - Attendance
 - Communication
 - Alerts
-

53. Web Applications

Business Admin Portal

- Dashboard
- CRM
- Sales
- Membership
- Finance
- Operations
- Trainers
- Inventory
- Marketing
- Support
- Reports

Trainer Portal

- Members
- Programs
- Assessments
- AI Copilot
- Sessions
- Progress

Platform Admin

- Tenants
 - Subscription
 - Configuration
 - AI models
 - Usage
 - Security
 - Audit
 - Support
-

54. Notification Engine

Central notification service supporting:

- Push
- WhatsApp
- Email
- SMS
- In-app

Trigger examples:

Membership expiry
Workout missed
Class reminder
Payment failure
Assessment due
Progress milestone
Trainer message
AI recommendation
Complaint update

55. Reporting & Audit

Every important business and AI operation should be auditable.

Reports:

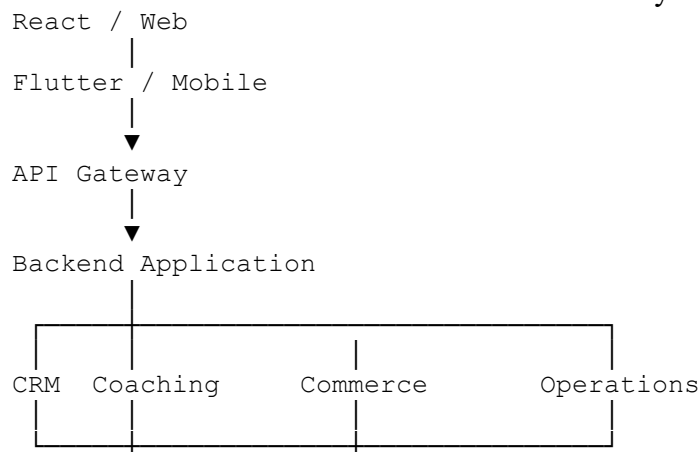
- Sales
- Membership
- Revenue
- Attendance
- Trainer
- Customer
- Workout
- AI usage
- CV usage
- Complaints
- Inventory
- Marketing

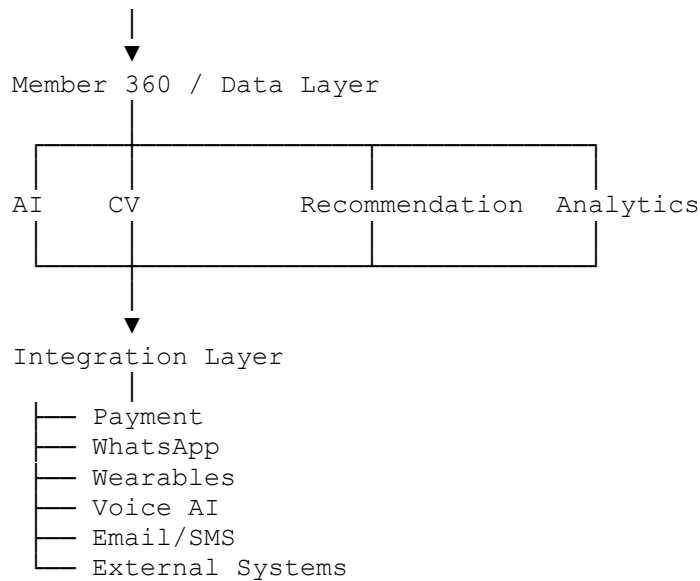
Audit:

- Login
 - Data access
 - Data modification
 - Export
 - AI recommendation
 - Permission changes
 - Consent changes
 - Administrative actions
-

56. Recommended Technical Architecture

The architecture should be modular even if initially deployed as a modular monolith.





For the backend, a **Django + PostgreSQL** foundation is appropriate for the business platform, with separate AI/CV inference services where required.

57. Development Principles

The developer must follow these principles.

1. API-first

Everything should be accessible through APIs.

2. Modular

CRM, Coaching, AI, CV, Commerce etc. should be independently extensible.

3. Multi-tenant

Tenant isolation from day one.

4. Configuration over hard-coding

Business rules should be configurable.

5. AI provider independent

Do not tightly couple the system to one LLM or AI vendor.

6. Human-in-the-loop

Especially for health-related coaching decisions.

7. Privacy by design

Especially for images, video, biometrics and health-related data.

8. Event-driven

Every meaningful member interaction should produce an event.

9. Explainable AI

AI should explain the reason behind significant recommendations.

10. Observability

Every AI and system component should have logs, metrics and tracing.

58. The Most Important Product Differentiator

The developer should understand that the product is **not**:

Gym Management + Chatbot.

It is:

A closed-loop AI fitness intelligence platform.

The loop is:

ASSESS
↓
PLAN
↓
TRAIN
↓
OBSERVE
↓
MEASURE
↓
UNDERSTAND
↓
ADAPT
↓
TRAIN AGAIN
↓
MEASURE OUTCOME
↓
LEARN

The more members use the platform, the more useful the **personalization engine** becomes—subject to appropriate consent, privacy controls, and responsible model development.

59. Final Product Modules

The developer's master backlog should therefore contain these modules:

A. Platform

1. Multi-Tenant Management
2. Organization
3. Branch
4. Role & Permission
5. Configuration
6. Subscription

B. Customer

7. Member Management
8. Member 360
9. Membership
10. Attendance
11. Booking
12. Customer Communication

C. CRM & Sales

13. CRM
14. Lead Management
15. Sales Pipeline
16. AI Sales Agent
17. AI Calling
18. Marketing
19. Campaign
20. Referral

D. Fitness

21. Assessment
22. Body Metrics
23. Fitness Score
24. Exercise Library

- 25. Workout Builder
- 26. Program Builder
- 27. Personal Training
- 28. Group Classes
- 29. Online Coaching
- 30. Progress Tracking

E. Nutrition & Wellness

- 31. Nutrition Profile
- 32. Food Database
- 33. Diet Generator
- 34. Meal Planner
- 35. Supplement
- 36. Sleep
- 37. Hydration
- 38. Activity
- 39. Lifestyle
- 40. Recovery

F. AI

- 41. AI Coach
- 42. Trainer Copilot
- 43. AI Content Generator
- 44. AI Workout Generator
- 45. AI Diet Generator
- 46. AI Recommendation Engine
- 47. AI Retention
- 48. AI Business Intelligence
- 49. AI Agent Framework
- 50. AI Audit Trail

G. Computer Vision

- 51. Pose Detection
- 52. Exercise Recognition
- 53. Rep Counting
- 54. Movement Analysis
- 55. Form Analysis
- 56. Person Tracking
- 57. Multi-Person Class Tracking
- 58. Real-Time Feedback
- 59. CV Analytics

H. Wearables

- 60. Wearable Integration
- 61. Health/Activity Data
- 62. Recovery Data
- 63. Device Management

I. Commerce

- 64. Product Catalogue
- 65. E-Commerce
- 66. Orders
- 67. Payments
- 68. Inventory

69. Vendor Management

70. Supply Chain

J. Operations

71. Center Operations

72. Room Management

73. Equipment

74. Trainer Management

75. Staff

76. Schedule

77. Facility Management

78. Safety

K. Customer Success

79. Grievance

80. Ticketing

81. Escalation

82. SLA

83. Feedback

84. CSAT

85. Customer Success

L. HR & Knowledge

86. HR

87. Trainer Certification

88. LMS

89. SOP

90. Knowledge Base

M. Analytics

91. Business Dashboard

92. Sales Analytics

93. Revenue Analytics

94. Member Analytics

95. Trainer Analytics

96. Fitness Analytics

97. AI Analytics

98. CV Analytics

99. Predictive Analytics

N. Security & Governance

100. Authentication

101. MFA

102. RBAC

103. Tenant Isolation

104. Consent

105. Privacy

106. Audit Logs

107. Data Retention

108. Data Export/Delete

109. AI Governance

110. Security Monitoring

60. Developer Definition of Done

A module should not be considered complete merely because the UI works.

Every module should include:

UI + API + Database + RBAC + Validation + Audit + Notifications + Error Handling + Reporting + API Documentation + Test Cases

For AI modules additionally:

Prompt/Model Version + Input Data Definition + Output Schema + Confidence/Guardrails + Human Approval + Audit Trail + Evaluation Dataset + Monitoring

For Computer Vision additionally:

Camera Input + Model + Inference + Latency + Person Tracking + Accuracy Measurement + Edge/Cloud Configuration + Privacy Controls + Failure Handling

Recommended Product Positioning

The final product should be positioned as:

AI Fitness & Wellness Operating System

For fitness centers, trainers and online coaches

Manage the business. Coach the member. Understand the body. Adapt the program.

Automate engagement.

The **business layer** gives your friend a practical commercial product today.

The **Member 360 + AI layer** creates the personalization engine.

The **Computer Vision + wearable layer** creates the long-term technical differentiation.

And the **event/data architecture** makes it possible to continuously improve the AI rather than rebuilding the product every time a new AI capability is introduced.

I would give this to the developer as the **Master Scope**, and then create a separate **module-wise PRD/API/database specification** from it. The next document should break these 110 modules into **Epic → Module → Feature → User Story → API → Database entities → Acceptance Criteria**, which will be much more useful for actual development than adding more conceptual features.